ASSISTMENTS 2009 · 313 TEST STUDENTS · KNOWLEDGE TRACING

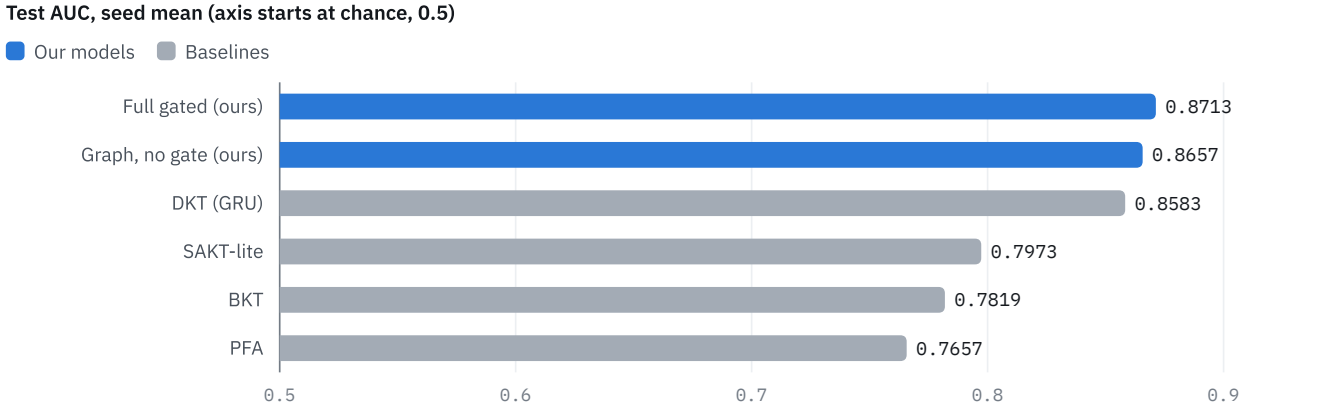
GatedKT comparison report

Results of the last three steps: the four-baseline benchmark, the 50-seed variance study, and hints and attempts inside the GRUs and the gate. Every number below is copied from a generated report in `reports/v3/`; the file is named under each section.

- The full gated model has the highest test AUC (0.8713) and improves over all four baselines: DKT (GRU), SAKT-lite, BKT and PFA.
- Over 50 seeds the gate's gain (+0.0062 to +0.0073 AUC) is 4–5× the seed-to-seed spread, and it wins on 50 of 50 matched seeds.
- Feeding hints and attempts into the GRUs as well as the gate is comparable to the gate alone (−0.0008 AUC). It changes predictions but not accuracy.

1. Against four baselines

The full gated model improves over every baseline (Holm-corrected Wilcoxon $p < 0.05$ and bootstrap CI of the AUC difference excludes 0). The closest is DKT (GRU), 0.0130 AUC behind.



Model	Seeds	Test AUC	95% CI (students)	Brier	Gain of full gated	Verdict
Full gated (ours)	3	0.8713	[0.8453, 0.8939]	0.1344	–	–
Graph, no gate (ours)	3	0.8657	[0.8392, 0.8898]	0.1370	+0.0056	▲ improves
DKT (GRU)	10	0.8583	[0.8316, 0.8826]	0.1408	+0.0130	▲ improves
SAKT-lite	10	0.7973	[0.7696, 0.8232]	0.1689	+0.0740	▲ improves
BKT	50	0.7819	[0.7407, 0.8187]	0.1715	+0.0894	▲ improves
PFA	50	0.7657	[0.7360, 0.7935]	0.1810	+0.1055	▲ improves

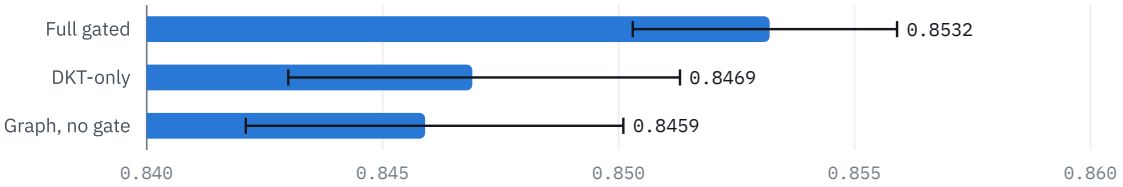
Source: reports/v3/benchmark.md. Our two models are the existing checkpoints (seeds 42, 123, 2024), trained earlier with a longer budget and extra inputs; the deep baselines had 10 epochs.

2. Is the gain bigger than seed noise?

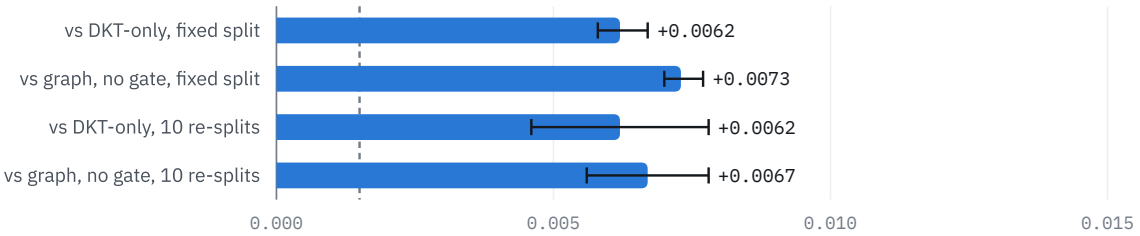
Fifty seeds per model on the fixed split, all retrained under the same budget. The gate's gain is far larger than the spread between seeds. When the student split also changes (10

splits), the spread grows ninefold but the gain stays.

Mean test AUC over 50 seeds, with min–max across seeds (fixed split)



Gain of full gated on matched seeds, with 95% CI



Dashed line: seed std of full gated on the fixed split (0.0015). Re-split std is 0.0138.

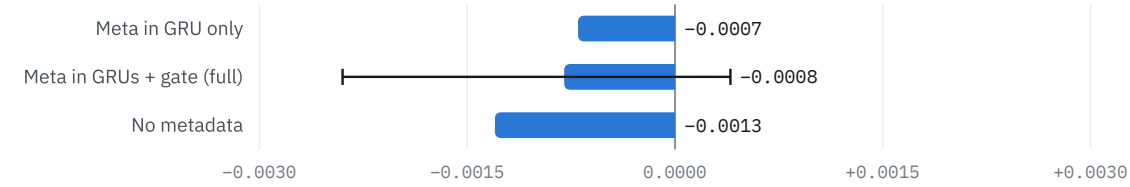
Model (fixed split)	Seeds	Mean AUC	Std	Min	Max	Gain of full gated [95% CI]	Verdict
Full gated	50	0.8532	0.0015	0.8503	0.8559	–	–
DKT-only	50	0.8469	0.0018	0.8430	0.8513	+0.0062 [+0.0058, +0.0067]	▲ improves
Graph, no gate	50	0.8459	0.0017	0.8421	0.8501	+0.0073 [+0.0070, +0.0077]	▲ improves

Source: reports/v3/seeds.md. Each run had 5 epochs with early stopping (15 epochs would have needed about 6.4 h), so these AUCs are lower than the checkpoints in section 1 and are compared only with each other.

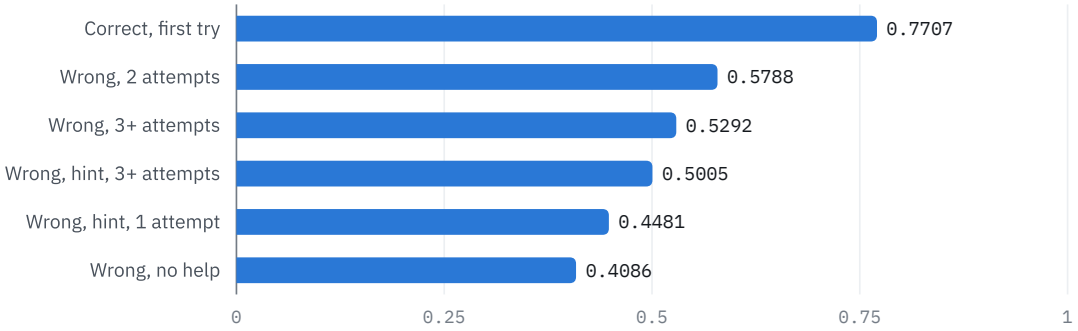
3. Hints and attempts in the GRU and the gate

GatedKT+ with hints and attempts in both GRUs and the gate is comparable to the version that feeds them to the gate only. The full version still improves over both DKT variants. The model does react to hints and retries: a wrong answer after two attempts lowers the next prediction much less than a plain wrong answer.

Test AUC difference from GatedKT+ (gate only), 10 seeds



Mean predicted P(next correct) after changing only the last answer (seed-0 model)



Model (10 seeds)	Hints and attempts enter	Test AUC	vs gate only [95% CI]	Verdict
GatedKT+ (gate only)	gate	0.8696	–	–
GatedKT+ meta (GRU)	GRU inputs	0.8689	−0.0007	–
GatedKT+ meta (full)	GRUs + gate	0.8687	−0.0008 [−0.0024, +0.0004]	= comparable
GatedKT+ (no metadata)	nowhere	0.8683	−0.0013	–
DKT classic	–	0.8613	–	full meta ▲ improves +0.0074
DKT (GRU)	–	0.8611	–	full meta ▲ improves +0.0076

Source: reports/v3/meta_assist.md (pre-registered in META_PREREG.md). Differences without a verdict are plain differences of means and were not separately tested against the gate-only model. The served app model is unchanged: the pre-registered serving rule failed (0.8696 < 0.8715).

Caveats

- Section 1 is not budget-matched: our checkpoints trained longer and also read response time, attempts and hints.
- Under an equal budget in the earlier Stage 5 benchmark, the full gated model was only comparable to classic DKT (reports/v3/stage5/benchmark.md).
- One dataset (ASSISTments 2009). Test-student bootstrap intervals (about ±0.025) are much wider than seed intervals.